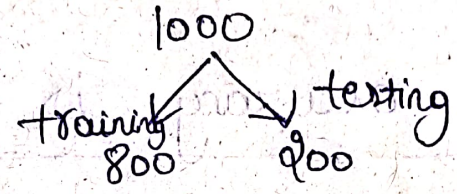
Evaluation for classification

Let's say im having 1000 Candidates data

cgpa	iq	placement



Actual	LgR	DTP
1	1 ✓	1 ✓
0	1 ✗	1 ✗
0	0 ✓	0 ✓
0	0 ✓	0 ✓
1	1 ✓	1 ✓
1	1 ✓	1 ✓
0	1 ✗	0 ✓
0	0 ✓	0 ✓
0	0 ✓	0 ✓
1	1 ✓	1 ✓

Accuracy:- $\frac{\text{Total no. of Correct}}{\text{Total predictions.}}$

$$LgR = \frac{8}{10} = 0.8 \Rightarrow 80\%$$

$$DTC = \frac{9}{10} = 0.9 \Rightarrow 90\%$$

Multiclass classification problem

Setosa — 0 | virginica — 1 | versicolor — 2

How much accuracy is good ?

→ it depends on problem we are Solving

ex :-

Cancer data

↓

96% → Can deploy ?

↓

if i give 100 patients it can correctly say 96 patients

Tesla Car

↓

98% → Can deploy ?

Swiggy

↓

89% → Can deploy ?

The problem with accuracy?

90% ⟶ 10% ✗

↓

don't explain about type of error.

Example Binary problem

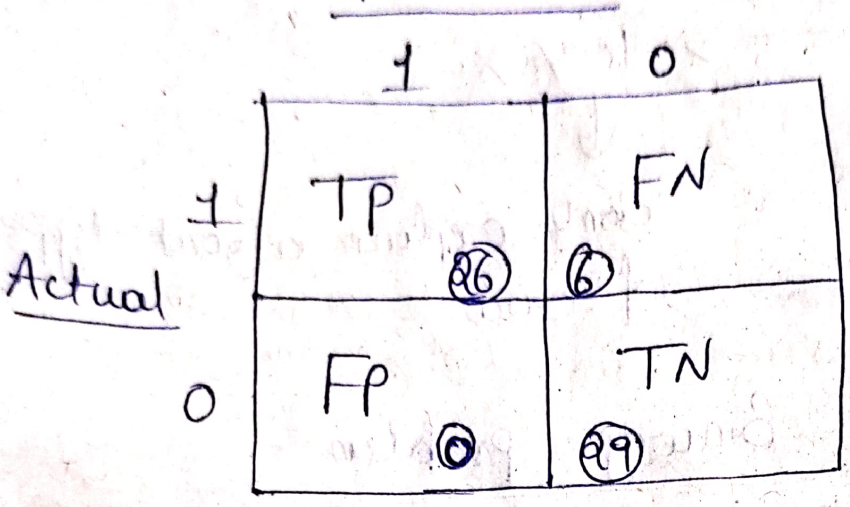
Y | N

→ Y ⟶ ✗

→ N ⟶ ✓

→ Your accuracy tell you what percentage of error is Caused but it didn't explain what type of error it is

→ That is why we use another metrics....



$$Acc = \frac{TP + TN}{TP + FN + Fp + TN}$$

Type 1 error :-

→ No. of Fp

Type 2 error :-

→ No. of FN

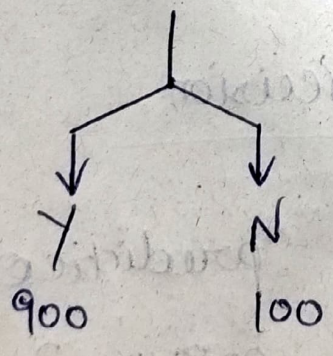
Confusion matrix for multiclassification problem

Predicted

Actual	0	1	2
0	7	0	5
1	2	21	6
2	9	1	13

→ when accuracy is misleading:-

→ Imbalanced dataset



[Airport model
100000
Create Confusion matrix & Caluclate accuracy.]

Precision :-

Manager 2 models

(A)

	Sent to SPAm	not Sent to SPAm
SPAm	100	170 FN
not SPAm	30	700
	FP	

(B)

	Sent to SPAm	not to SPAm	
SPAm	100	190	FN
not SPAm	10	700	
	FP		

→ Because Both accuray's are Same.

Model A FP > Fp of B

F_N A < F_N B

This is Called precision

(i) what propotion of predicticed positives are truly positive?

Precision $\Rightarrow \frac{TP}{TP+FP}$

$\Rightarrow$ Ratio of TP in whole positive predicted

Precision of model A:-

model A $\Rightarrow \frac{TP}{TP+fp} \Rightarrow \frac{100}{100+30} \Rightarrow \frac{10}{13}$

Model B $\Rightarrow \frac{100}{100+10}$

$\therefore$ Model A < model B

Recall :-

Cancer

Model A

	Detected cancer	Not Detected	
Has Cancer	1000	200	FN
No Cancer	800	8000	
	Fp		

Model B

	Detected Cancer	Not detected	
Has Cancer	1000	500	FN
No Cancer	500	8000	
	Fp		

→ Again accuracy for both models are same

Recall :-

what proportion of actual positive are correctly classified?

formula :- $\frac{Tp}{Tp+FN}$

Model A $\Rightarrow \frac{1000}{1000+200} = \frac{1000}{1200}$ ↑

Model B $\Rightarrow \frac{1000}{1000+500} = \frac{1000}{1500}$ ↓

~~$M_B > M_A$~~

$\therefore$ MA

F1 Score :-

→ Imagine a Dog/Cat classification model, we can't decide wether 'precision/recall

formula :- $\frac{2PR}{P+R}$.

Note :- we don't use arthematic mean we use harmonic mean.

But why?

↳ By using harmonic mean we penalize the model with lower precision / Recall

Ex 1:-

if model with

$P = 0$, Recall $= 100$.

what is arth mean $\Rightarrow \frac{100}{2} = 50$.

Harmonic mean? $\Rightarrow 0$.

Model (A)

$P = R = 80$

$\text{Arth} = \frac{80+80}{2} \Rightarrow 80$

Model B

$P = 60$, $R = 100$.

$\text{Arth} \Rightarrow \frac{60+100}{2} \Rightarrow 80$.

$A \Rightarrow \frac{2 \times 80 \times 80}{160} \Rightarrow 80$

$B \Rightarrow \frac{2 \times 60 \times 100}{160} \Rightarrow 75\%$

↳ precision is low so it is penalizing.

F1 score takes stand of lower values.

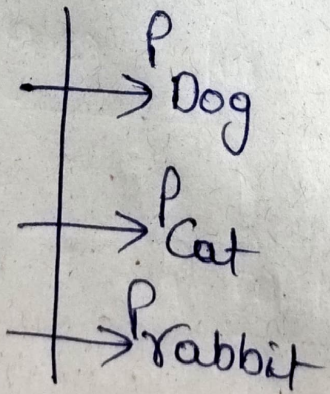
Multiclass precision recall f1 score

→ In Binary classification we only focus on "1" class.

→ Behind the scence we calculate Both but only pick "1"

average = None

→ Now we have to calc precision



$P_{dog} \Rightarrow \frac{25}{29} \Rightarrow 0.86$

$P_{cat} \Rightarrow \frac{30}{45} \Rightarrow 0.66$

$P_{rabbit} \Rightarrow \frac{20}{34} \Rightarrow 0.58$

→ Now we need to Calc precision.

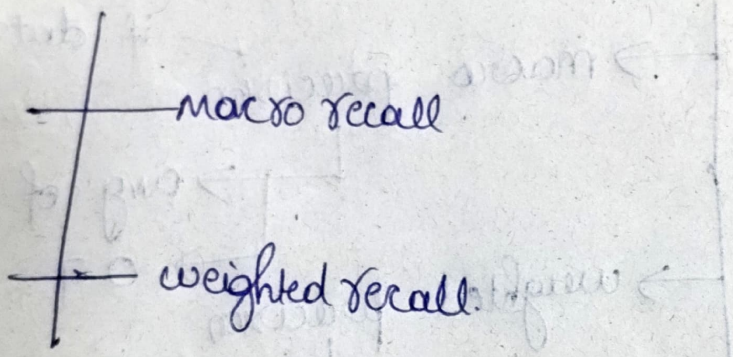
- macro precision ← if data is balanced
  - avg of all precision
  - 0.7
- weighted precision
  - $\frac{40}{108}$ 0.86 + $\frac{34}{108}$ 0.66 + 0.58 $\frac{34}{108}$
  - ⇒ 0.71

Recall :-

R_D , R_C, R_R

$R_D = \frac{25}{40}$, $R_C = \frac{30}{34}$, $R_R = \frac{20}{34}$

$\Rightarrow 0.62$, $\Rightarrow 0.88$, $\Rightarrow 0.58$



∴ Same goes for F_1 score :-

$$F_{1D} = \frac{2 P_D R_D}{P_D R_D}$$